

# Patient, Neighborhood, and Spatial Determinants of Out-of-Hospital Cardiac Arrest Outcomes Throughout the Chain Of Survival: A Community-Oriented Multilevel Analysis

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## INTRODUCTION

### Background:

- Death and disability from out-of-hospital cardiac arrest (OHCA) remains a prominent public health threat in the United States (350,000 deaths annually)
- Overall survival rate to hospital discharge (12%) has remained stable over time
- National disparities have been examined, where the survival to discharge rate has been observed to be less than 6% in some areas
- Geographic disparities in the community setting are less understood

### Objective:

- We investigated the spatial, patient, and neighborhood factors that are associated with key OHCA survival milestones in Alachua County, FL

### Public Health Relevance

- This precision public health surveillance research demonstrates the current status of cardiac arrest outcomes in our community while identifying local areas where critical modifiable interventions can be improved.

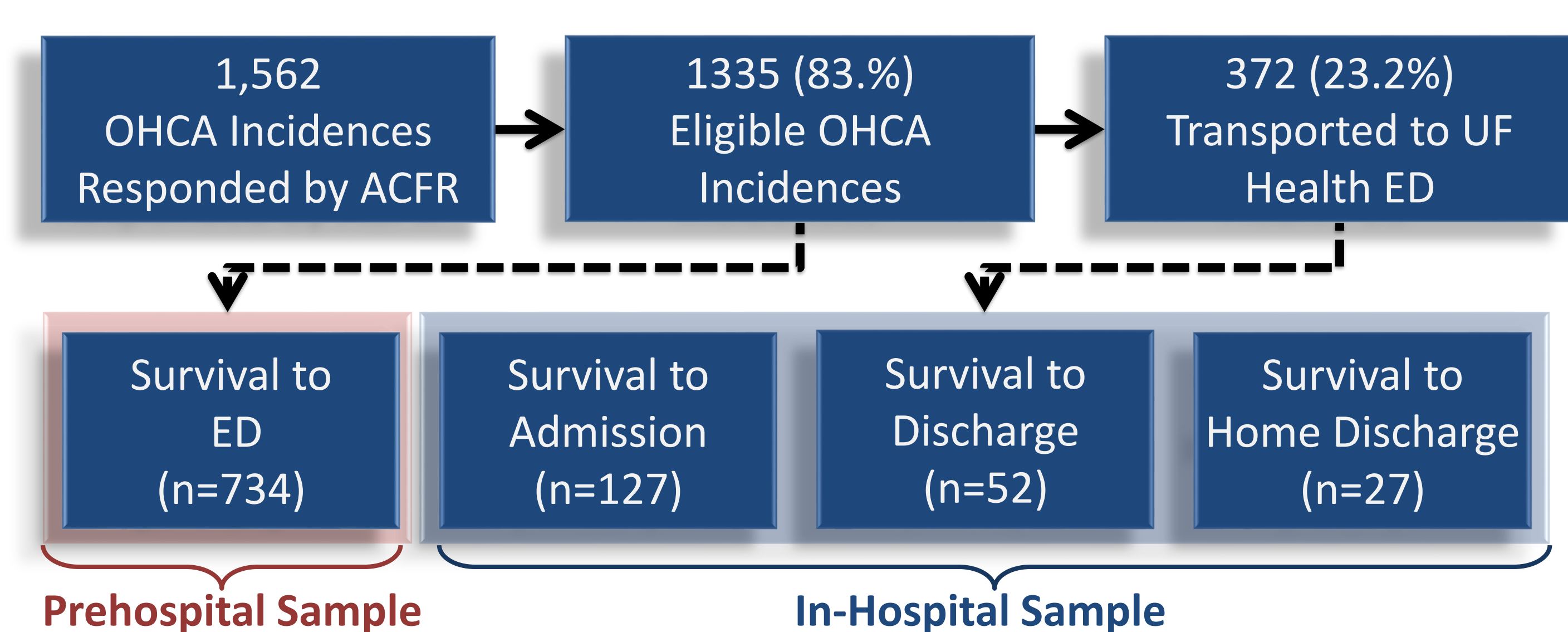
## METHODS

### Data Overview & Processing

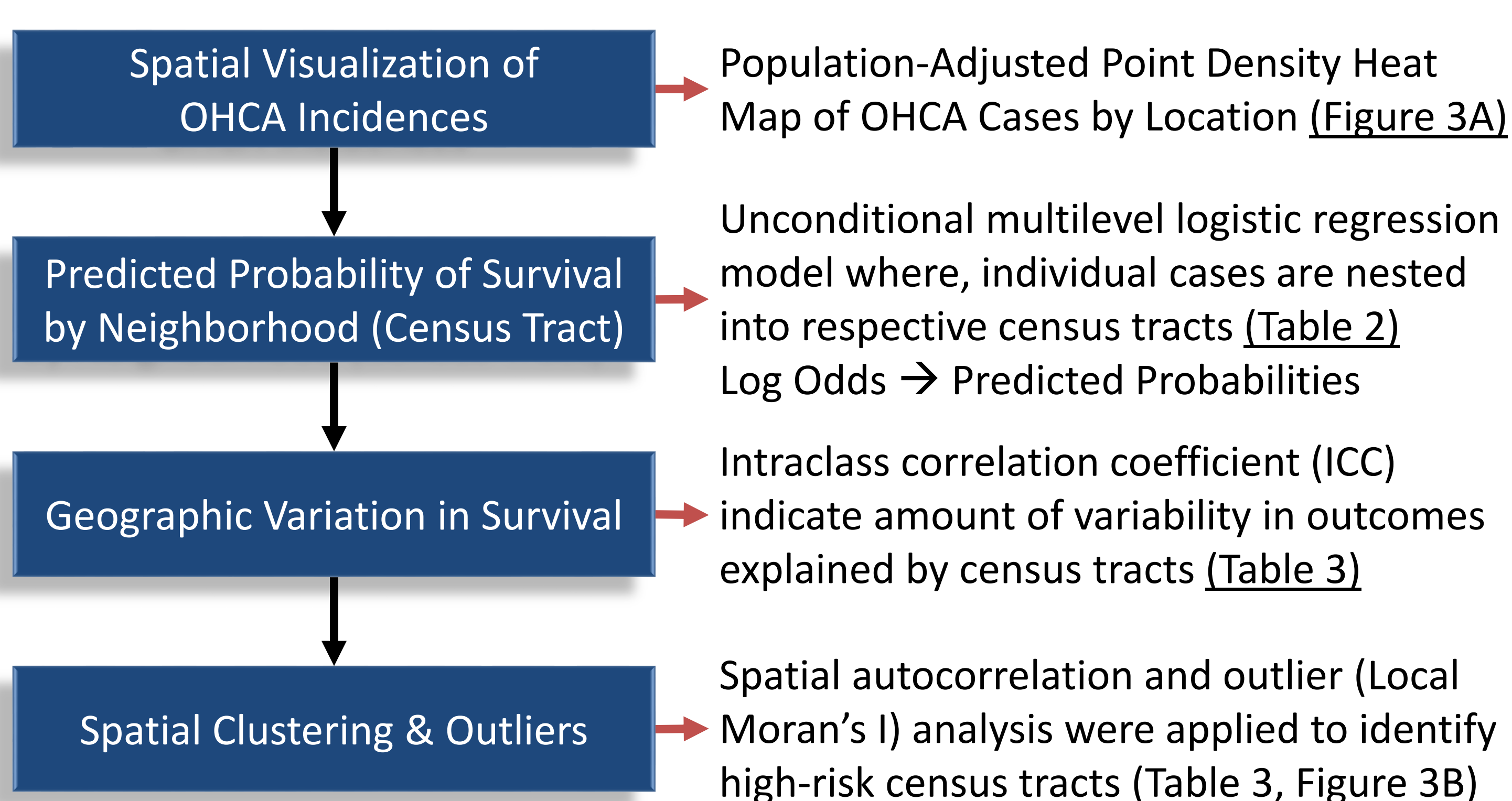
- Alachua County Fire Rescue (ACFR) EMS records of non-traumatic, adult OHCA between 01/2012 and 01/2017 were retrospectively reviewed.
- ACFR records were individually matched to corresponding University of Florida electronic medical record via UF Health Decision Support Services (DSS).
- Location of cardiac arrest incidence was geocoded and was spatially joined to corresponding census tract (ArcGIS).

### Primary Outcomes (4)

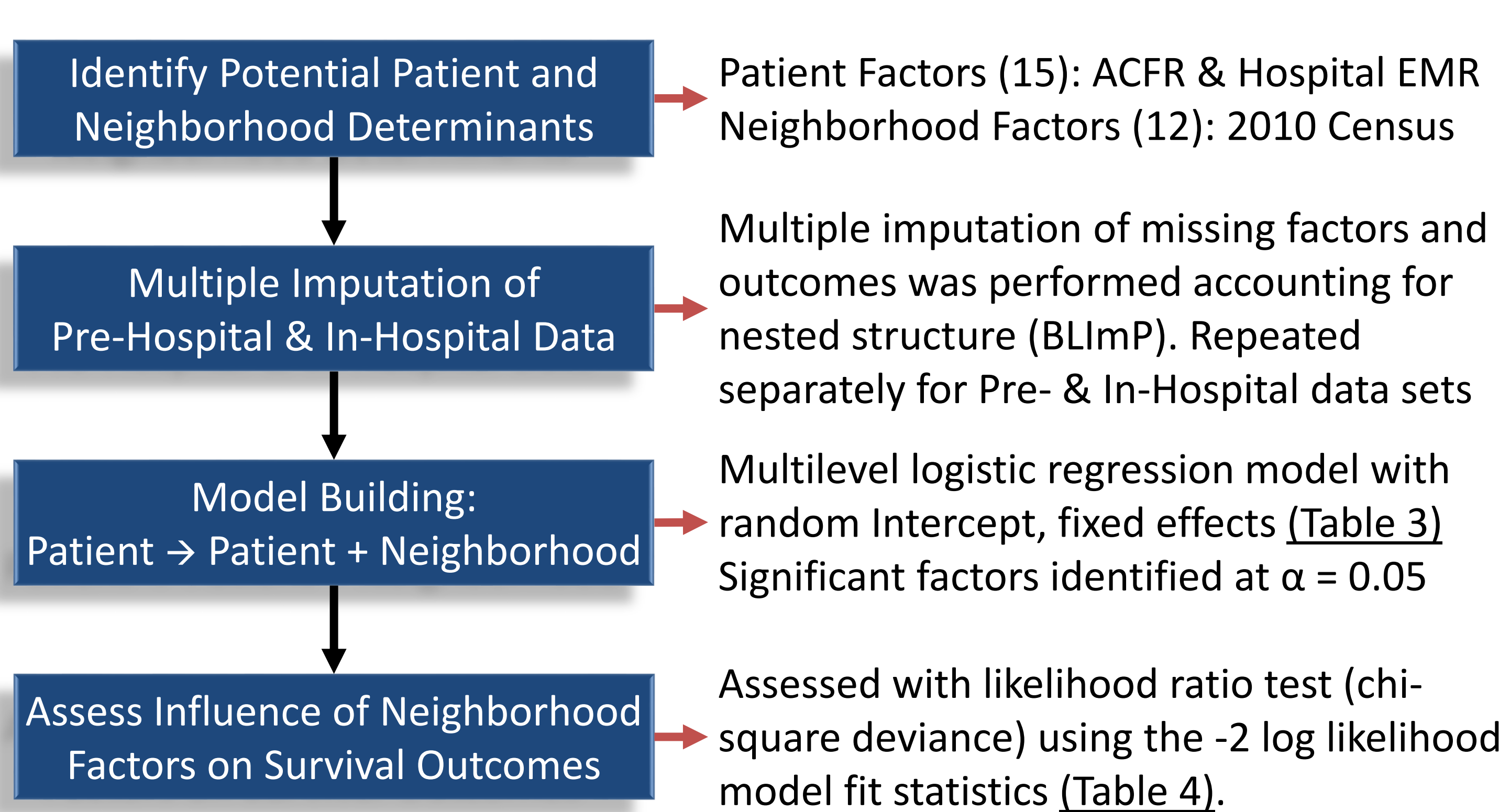
- Survival to Emergency Department
- Survival to Hospital Admission
- Survival to Hospital Discharge
- Survival to Home Discharge



### Multilevel Modeling (1) – Survival Probabilities & Geographic Variation



### Multilevel Modeling (2) – Patient & Neighborhood Determinants



## SURVIVAL PROBABILITIES & GEOGRAPHIC VARIATION

**Table 1:** Study demographics of eligible OHCA incidences responded by ACFR in time period

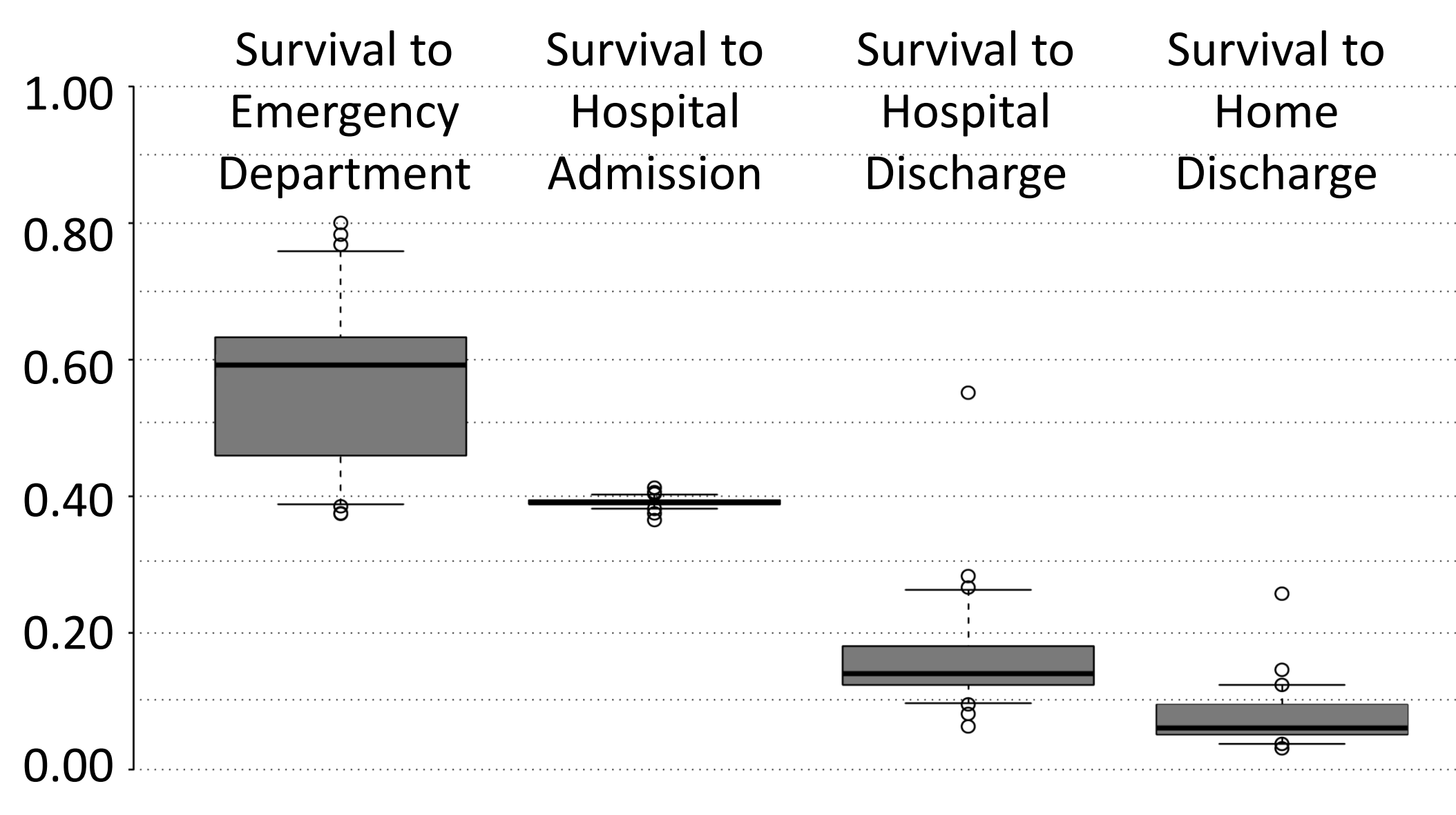
| Patient Predictors           | Study Population (n = 1335) |
|------------------------------|-----------------------------|
| Age, mean (SD)               | 66.3 (16.9)                 |
| Sex (% Female)               | 541 (40.5)                  |
| Race:                        |                             |
| Black/African-American       | 350 (26.2)                  |
| White                        | 967 (72.4)                  |
| Other/Unknown                | 18 (1.4)                    |
| Witnessed Arrest:            | 540 (41.7)                  |
| ROSC Prior to ED Arrival:    | 271 (20.3)                  |
| Bystander AED:               | 68 (7.3)                    |
| Bystander CPR:               | 400 (42.9)                  |
| Arrest at Home (vs. Public): | 916 (68.6)                  |

**Table 2:** Estimated probabilities of survival outcomes among prehospital (red) and in-hospital (blue) samples

| Survival Outcome   | Probability (95%CI)   |
|--------------------|-----------------------|
| Emergency Dept.    | 57.0% (51.3% - 62.6%) |
| Hospital Admission | 41.7% (36.1% - 47.6%) |
| Discharge          | 16.1% (10.7% - 23.5%) |
| Home Discharge     | 7.1% (3.7% - 13.3%)   |

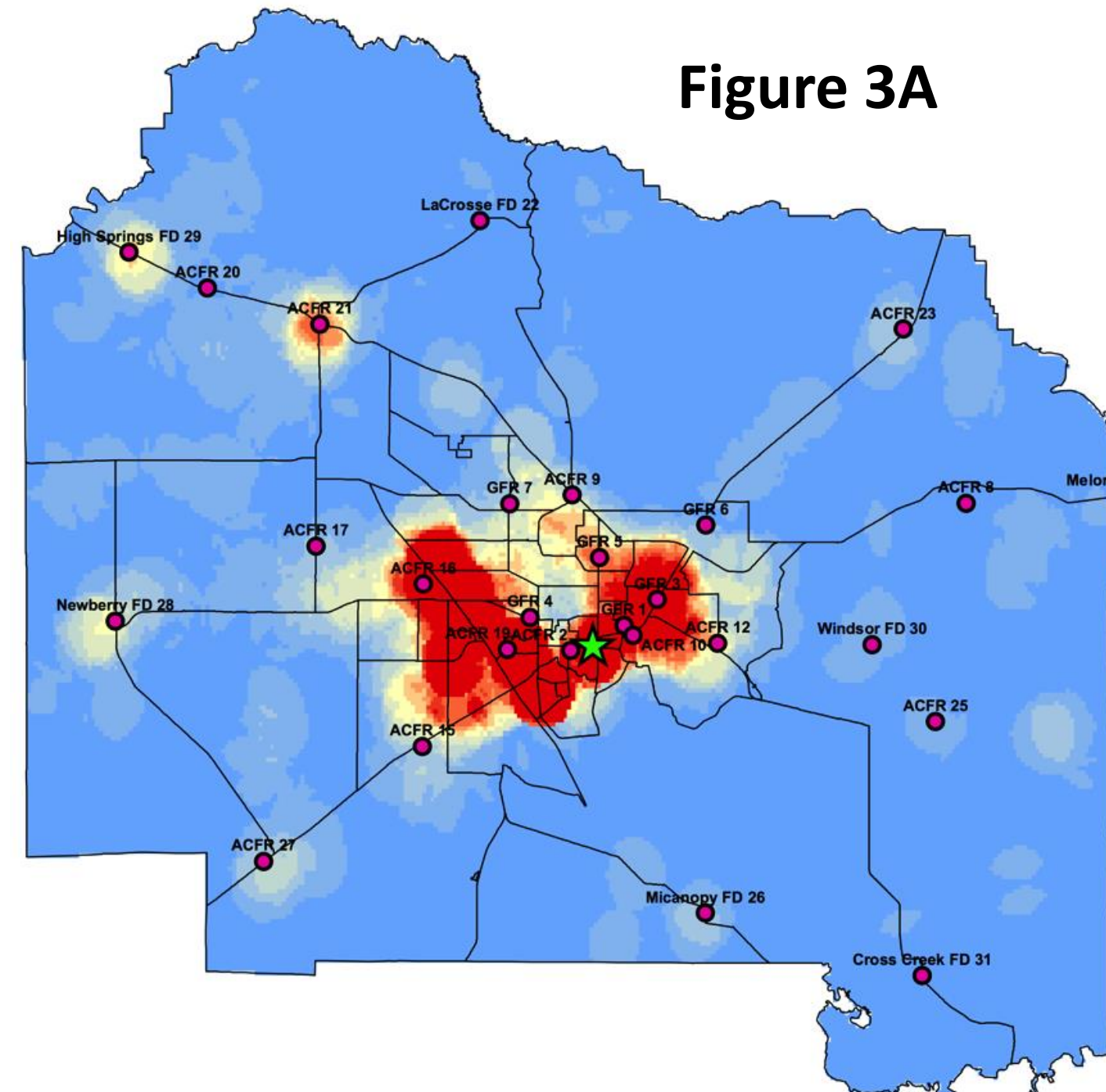
**Table 3:** Intraclass correlation (ICC) and spatial clustering analysis (Moran's I) of survival outcomes of study population

| Survival Outcome   | ICC (P value)  | Moran's I (P value) |
|--------------------|----------------|---------------------|
| Emergency Dept.    | 12.2% (<0.001) | 0.080 (<0.001)      |
| Hospital Admission | 1.5% (0.312)   | -0.019 (0.765)      |
| Discharge          | 18.0% (0.031)  | -0.019 (0.774)      |
| Home Discharge     | 26.0% (0.036)  | 0.049 (0.099)       |

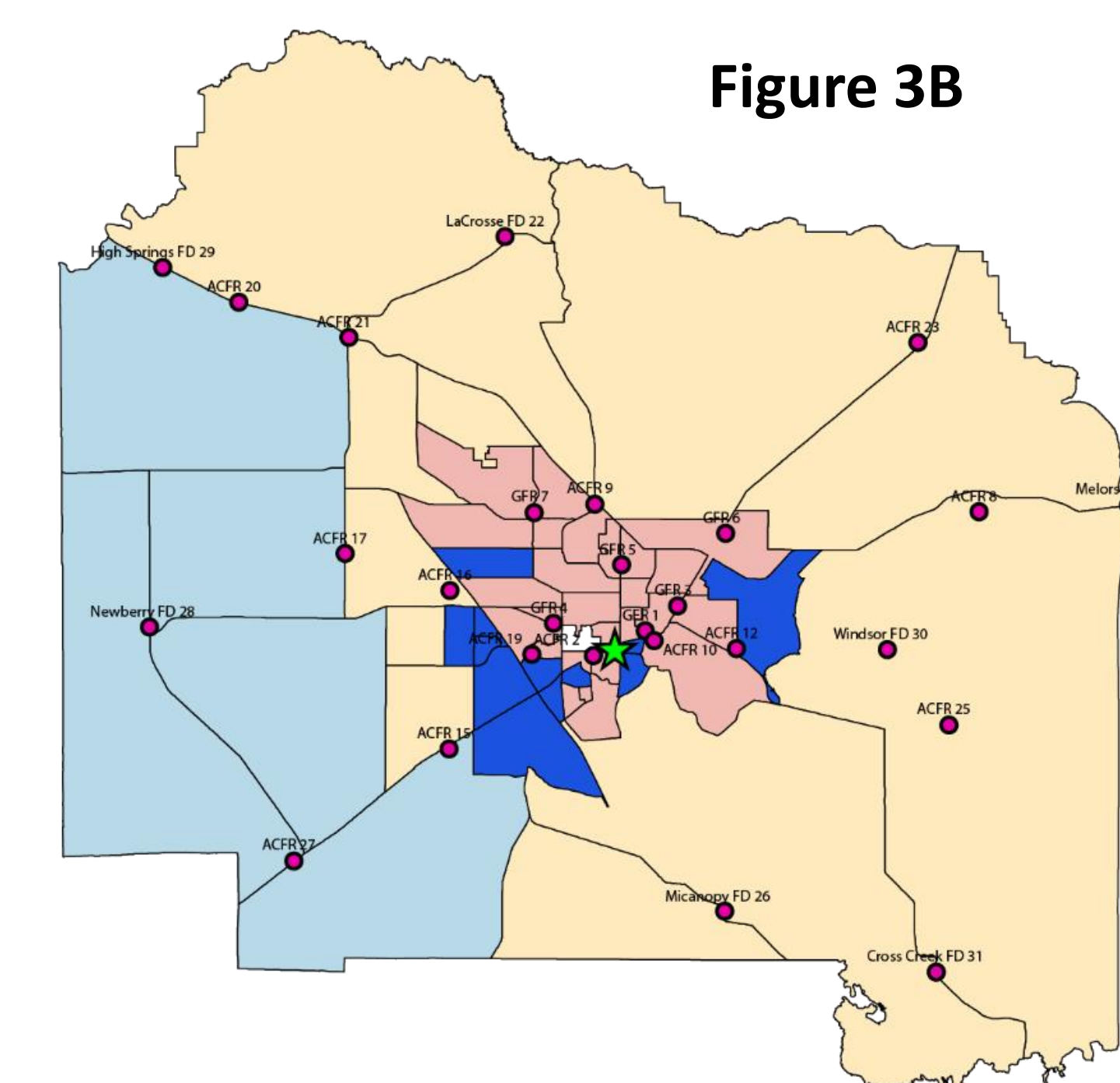


**Figure 1:** Box plots of predicted probabilities of survival outcomes by census tract (n=45)

**Figure 3A:** Point density map of population-adjusted annual OHCA density (OHCA/sq. km/year) [Range: 0 – 60]



**Figure 3B:** Outlier spatial analysis of predicted probabilities of survival to emergency dept. by census tract (n=45)



## PATIENT & NEIGHBORHOOD DETERMINANTS

**Table 3:** Significant determinants obtained from final adjusted models (patient + neighborhood effects) for each survival outcome \*Indicates neighborhood factor

| Negative Survival Associations | Positive Survival Associations |
|--------------------------------|--------------------------------|
| <b>Emergency Dept.</b>         | <b>Emergency Dept.</b>         |
| Age                            | Black/African-American         |
| Cardiac Arrest At Home         | Witnessed Arrest               |
| Arrest Before EMS Arrival      | Bystander CPR                  |
| Percent Vacant Housing*        | Shockable Initial Rhythm       |
| Rural Census Tract*            | Arrest in Afternoon-Evening    |
|                                | Time EMS Dispatch-Scene        |
| <b>Hospital Admission</b>      | <b>Hospital Admission</b>      |
| Age                            | ROSC Prior to ED Arrival       |
|                                | Shockable Initial Rhythm       |
| <b>Hospital Discharge</b>      | <b>Hospital Discharge</b>      |
| Age                            | ROSC Prior to ED Arrival       |
| Time EMS On Scene              | Shockable Initial Rhythm       |
| Unemployment Rate*             |                                |
| <b>Hospital Home Discharge</b> | <b>Hospital Home Discharge</b> |
| Age                            | ROSC Prior to ED Arrival       |
| Percent College Graduate*      | Bystander AED Use              |

**Table 4:** Assessment for improvement of model fit after the addition of patient- and neighborhood-level factors using -2 log likelihood statistics.

\* Indicates statistically significant improvement

**Model 1:** Unconditional

**Model 2:** Patient-Level Predictors ONLY

**Model 3:** Patient + Neighborhood Predictors

| Survival Outcome   | Model 1 | Model 2 | Model 3 |
|--------------------|---------|---------|---------|
| Emergency Dept.    | 1778.7  | 1093.5* | 1056.6* |
| Hospital Admission | 504.3   | 339.5*  | 319.6   |
| Discharge          | 340.9   | 202.8*  | 176.4*  |
| Home Discharge     | 227.3   | 117.9*  | 89.0*   |

## CONCLUSION

- Geographic disparities in OHCA outcomes exist in the sub-county context, including survival measured at hospital discharge
- Variation is attributable to prehospital and neighborhood socioeconomic factors
- High-risk census tracts are identified
- Provides the basis for active community-based participatory programs targeting bystander intervention (e.g. PulsePoint)